

Program : 4

Manipulate the HTML DOM using JavaScript

Aim: To Manipulate the HTML DOM using JavaScript.

Algorithm:

Step 1: Initialize the Webpage

- a. Load the HTML structure with a heading (`h1`) and a button (`button`)
- b. Assign an `id` to the heading (`myHeading`) and button (`changeButton`).
- c. Apply initial CSS styles, including transition effects for smooth visual changes.

Step 2: Access HTML Elements

- a. Use `document.getElementById("myHeading")` to store the heading element.
- b. Use `document.getElementById("changeButton")` to store the button element.

Step 3: Add an Event Listener to the Button

- a. Attach a `click` event listener to the button.
- b. When the button is clicked:
 - Update the heading text to "Text has been changed!".
 - Toggle the heading color between **blue** and **red**.
 - Toggle the font size between **32px** and **24px**.

Step 4: Apply Changes Dynamically

- a. Modify the `textContent` property of the heading to update its text.
- b. Use conditional checks (`if-else` or ternary operator) to toggle the color and font size.
- c. The transition effect ensures a smooth change.

Step 5: Repeat on Every Click

- a. Every time the button is clicked, the text, color, and font size toggle dynamically.
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Source Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>DOM Manipulation</title>

</head>

<body>

  <body>

    <h1 id="myHeading">welcome to HTML DOM in Java Script Page</h1>

    <button id="changeButton">Change Text and color</button>


    <script>

      const heading = document.getElementById("myHeading"); // Get the heading element

      const button = document.getElementById("changeButton"); // Get the button element


      button.addEventListener("click", function() {

        heading.textContent = "Text has been changed!"; // Update the heading text

        heading.style.color = "blue"; // Change the heading color

      });

    </script>

  </body>

</html>
```

Output:

Before Running:

**welcome to HTML DOM in Java
Script Page**

Change Text and color

After Running:

Text has been changed!

Change Text and color